# FoveaWAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

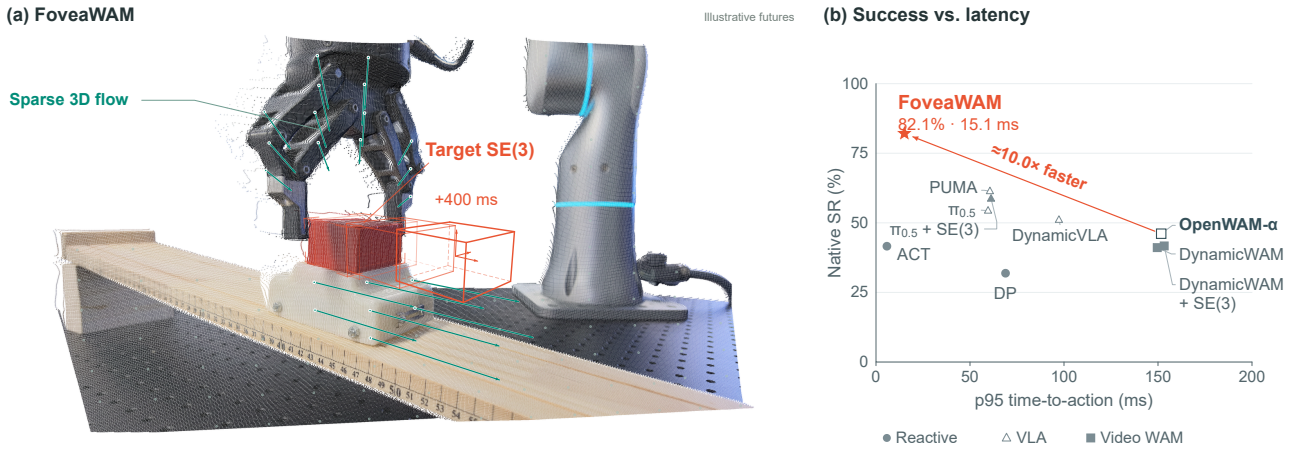


Figure 1. **FoveaWAM allocates prediction precision by task role.** (a) A real-robot scene reconstructed from RGB with MoGe-2 [42], with illustrative target SE(3) futures and sparse 3D flow. (b) Native SR versus p95 time-to-action from Table 1.

## Abstract

*To manipulate a moving object, a policy must predict the scene at the moment its action executes, yet every millisecond spent predicting pushes that moment further away. Existing policies either forecast the whole scene at uniform fidelity and act late, or act fast by ignoring everything but the target. Our key insight is that what a policy must predict decouples by role: only the target's pose, which decides contact, needs millimeter precision, while the surroundings matter only through the clearance they leave and the motion they impart. We present FoveaWAM, a native 4D world–action model that allocates precision by role: it models the target's past and future motion as a metric SE(3) trajectory and the surroundings as sparse 3D flow, and relies on the current frame for everything static. This compact representation lets FoveaWAM act in 15.1 ms on one RTX 5090, about 10× faster than video world–action baselines, giving the scene little time to move. To evaluate each policy under its own latency, we introduce OnTimeBench, a dynamic manipulation benchmark in which the scene keeps moving while the policy computes. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, and also leads on DOMINO and real-robot tasks.*

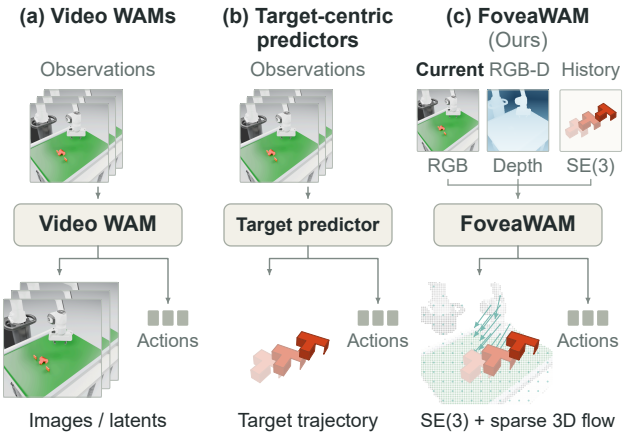


Figure 2. **What dynamic policies forecast.** (a) Video WAMs forecast future frames or latents at uniform fidelity [43, 50]. (b) Target-centric predictors forecast only the target's trajectory [13, 16]. (c) FoveaWAM reads one RGB-D frame with the target's SE(3) history and forecasts the target's SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow; both forecasts supervise training and are skipped at inference.

## 1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [5, 6, 11, 20, 43, 50, 64], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. In such scenes, an action must fit the scene at the moment it executes, not the moment its observation was captured, so a policy has to know how the scene moves. That knowledge must itself be inferred, and inference takes time during which the scene keeps moving: the longer a policy computes to anticipate the motion, the further ahead it must anticipate.

Recent dynamic policies add a memory of past motion, typically image-space cues such as optical flow [13, 27], and a forecast of the future [27, 38, 57]. Video-based WAMs forecast whole future frames or latents at uniform fidelity, spending much of their computation on visual detail [43, 50] (Fig. 2a). Compact future codes [28] reduce this cost and metric 3D scene dynamics [55] add geometry, but both still treat the whole scene uniformly. In each case, the quantity that decides contact, the target's metric pose at execution time, stays implicit. Target-centric predictors and dynamic VLAs with object-level forecasts sit at the other end [13, 16] (Fig. 2b): they are fast but anticipate only the target, leaving out the surroundings that can block or redirect its motion. Neither approach allocates precision according to what contact requires from each part of the scene. This raises a central question: what must a policy know about a moving scene, and how precisely, to make contact while keeping inference latency low?

Our key observation is that a dynamic policy needs to know very little precisely (Fig. 1). Contact demands millimeter precision in the target's pose at execution time; the surroundings matter only through the clearance they leave and the motion they impart to the target, so they need only be known coarsely. Building on this observation, we introduce FoveaWAM, a world–action model that learns a role-aware metric 4D representation, precise where contact demands it (Fig. 2c). To supply motion information without encoding past image frames, FoveaWAM pairs a single RGB-D frame with a short history of the target's metric SE(3) poses. A world stream learns the representation by forecasting the target's future SE(3) trajectory and the motion of the whole scene as a sparse 3D flow. An action stream reads the world stream's scene and summary tokens to generate action chunks, with explicit forecasting confined to training [54]. FoveaWAM thus combines precise anticipation of target motion with low inference latency, limiting how much the scene changes between observation and action.

We build OnTimeBench, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference, so every method pays for its own latency. FoveaWAM achieves 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, at a p95 latency of 15.1 ms with compiled BF16 inference, about 10× lower than that of the video world–action baselines [27, 43]. It reaches 46.8% on 24 DOMINO tasks [13], 19.9 points above DynamicWAM, and succeeds in 37.5% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Controlled ablations test the contributions of target-motion history and future supervision, and compare role-aware with uniform-fidelity forecasts to examine the allocation of precision.

Our contributions are threefold:

- We show that a dynamic policy needs millimeter precision only in the target's pose at execution time, and instantiate this principle as role-aware 4D modeling: a precise metric SE(3) trajectory for the target and a coarse sparse 3D flow for the whole scene.
- We develop FoveaWAM, a world–action model that combines metric target-pose histories with role-aware predictive supervision for low-latency action generation.
- We introduce OnTimeBench, an open-source dynamic manipulation benchmark that charges each policy its own inference latency, and demonstrate gains over strong baselines on OnTimeBench, 24 DOMINO tasks, and six real-robot tasks.

## 2. Related Work

**Dynamic manipulation.** Compensating for the time between sensing and acting is a classical problem, from control with dead time [35] to catching systems that track the target with Kalman-type filters and plan an interception [1–3, 30]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes [48]. PUMA reads historical optical flow and is supervised with future object features [13], and DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [27]. Both infer past motion from image-space cues, whereas FoveaWAM takes the target's metric SE(3) pose history as input and observes only the current RGB-D frame. Other dynamic policies forecast salient tokens [38], future images [58], compact future codes [28], or 3D scene dynamics [55] and condition their actions on these forecasts at test time. DynaWM adapts a base VLA's action chunks for dynamic manipulation [10], and DynamicManip synthesizes dynamic demonstrations from a static one [22]. Execution schemes reduce or hide latency by overlapping inference with execution [33, 48], keeping consecutive chunks consistent [7, 25], or conditioning on the execution time [39]. They decide when actions are computed and applied, not what the policy represents. DOMINO [13] and PhysMani-Bench [55] provide diverse

dynamic-manipulation tasks, but their released default evaluation pipelines advance the simulation only after a policy has returned its prediction. OnTimeBench advances the scene during inference and charges each method its measured latency.

**Predictive supervision in WAMs.** World–action models (WAMs) generate future video and actions in one network [21, 43, 50, 62], building on video prediction as a policy objective [40, 47]. Because generating video is costly, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [57], flow and geometry alongside RGB [53], object masks or slots [26, 51], or compact latents [37, 61], and asks whether the forecast is needed at test time. Fast-WAM retains most of the benefit of video modeling without test-time imagination [54], and PUMA and SG-WAM drop their predictors after training [13, 59]. FoveaWAM keeps this training-only use but forecasts primarily a metric 4D state rather than pixels, features, or latents, so the target's future pose is supervised explicitly instead of being left implicit.

**Object-centric and geometric motion representations.** Metric 3D inputs make policies more precise and sample-efficient [32, 56]. Motion has been represented by 2D point tracks [4, 46, 49], 3D flow [14, 17, 52, 55], and object keypoints or SE(3) trajectories [16, 29, 36], while rigid-body decompositions have long been used to learn scene dynamics [8, 9]. These works describe the scene with one kind of representation, such as dense flow or tracks everywhere, or a pose for each object. Target-centric predictors, including dynamic VLAs with object-level forecasts, model only the target's future [13, 16] and leave out the bodies that block or push it. DOCIR separates the robot, the target, and obstacles, but uses these distinctions only as input encodings [12]. FoveaWAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, while the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast.

# 3. FoveaWAM

We formulate dynamic manipulation as generating time-indexed command chunks (Sec. 3.1) from a representation supervised, at the execution times of these commands, with the scene's sparse 3D flow and the target's SE(3) trajectory (Fig. 3). We describe the metric inputs (Sec. 3.2), this supervision (Sec. 3.3), and how an action stream reads the representation to generate the chunk (Sec. 3.4).

## 3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query $t$, a policy $\pi$ learned from demonstrations reads an RGB image $I_t$ and a depth map $D_t$ captured at time $\tau_t$, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of $H$ commands

$$(\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command $j$ is meant for time $\tau_t + j\Delta_a$, with control period $\Delta_a$.

Each task specifies a *target*, whose pose the robot must match within a contact or containment tolerance. The surroundings that matter form the *context*: obstacles, which the robot must clear, and drivers, which move the target. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at $\tau_t$.

## 3.2. Metric Scene and Motion Inputs

FoveaWAM encodes the target's motion precisely and its context through the current geometry: the current RGB-D frame gives the metric layout of the scene, and a short history of target poses gives the motion that one frame does not show, which also reflects how drivers have moved the target.

**Current scene.** FoveaWAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the forecasts and the actions. A frozen MAE ViT-B/16 [15] turns the RGB image into 196 patch tokens on a $14 \times 14$ grid, and the depth map back-projects the center of patch $i$ to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

**Target motion.** Instead of past frames, from which motion must be recovered by comparing images, FoveaWAM receives the target's recent motion as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units. The velocity and acceleration that decide where the target will be thus enter the model directly, at the cost of a few low-dimensional tokens; Sec. 4.3 removes this history to measure its effect. In simulation, we use ground-truth target poses from the simulator. On the robot, a causal tracker estimates them from the RGB-D stream.

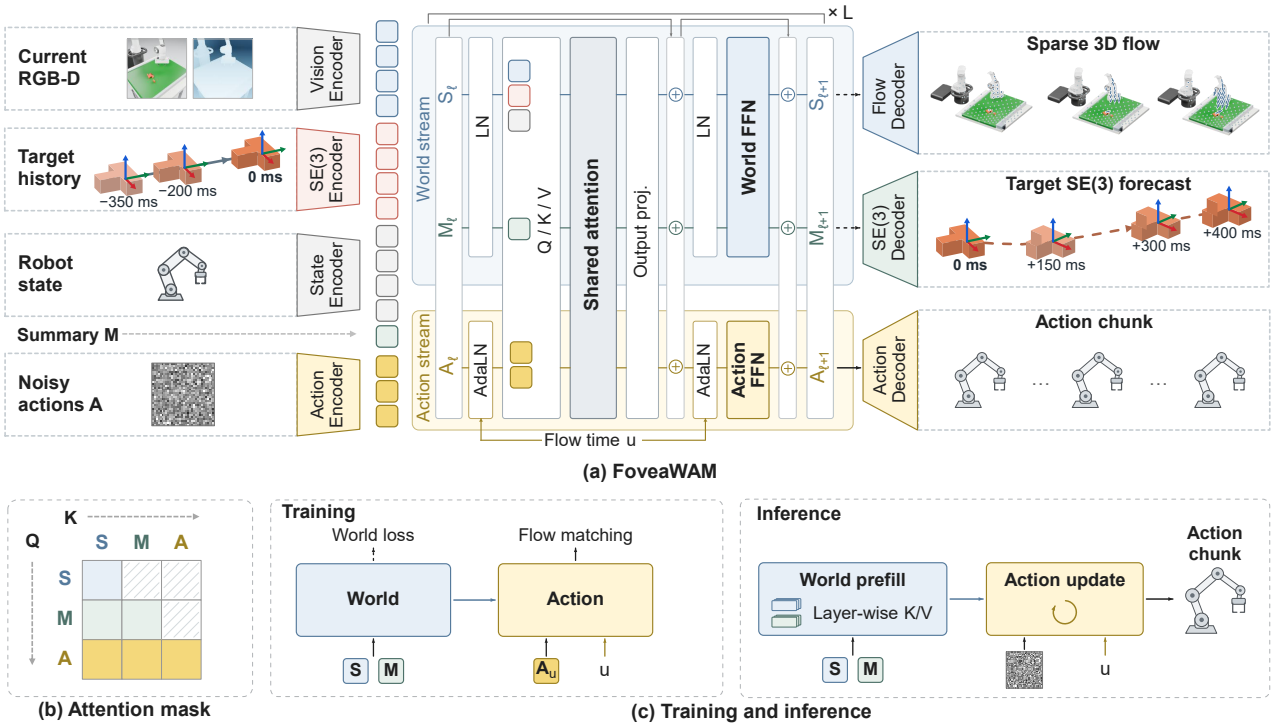


Figure 3. **FoveaWAM architecture.** (a) Current RGB-D and target/robot histories form scene tokens $S$, alongside learned summaries $M$ and noisy actions $A$. Each layer shares masked attention, applies one World FFN to $S$, $M$ and a separate Action FFN to $A$, and retains both residual updates. Final patch and mean-summary states support training-only geometric readouts (dashed); actions read hidden states. (b) Mask rows are queries and columns are keys. (c) Joint training combines auxiliary supervision and action flow matching. At deployment, world prefill caches layer-wise keys and values for action sampling, skipping geometric readouts. The RGB thumbnail is a Blender rendering. The target history (solid) and future SE(3) trajectory (dashed) meet at the current pose (0 ms). The future trajectory and sparse flow illustrate supervision from the recorded rollout (flow enlarged for display). Token counts, layer stacking and action icons are schematic.

**World stream.** The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream.

### 3.3. Role-Aware 4D Predictive Supervision

FoveaWAM supervises its representation with two geometric forecasts of different precision: the motion of the whole scene as a sparse 3D flow on the patch grid, and, because the target's pose decides contact, the full SE(3) trajectory of the target. Both are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \ldots, 8$, from 50 ms to 400 ms after capture; together with the current state, these horizons cover the execution times of all commands in the chunk.

**Scene flow.** At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for. For each of the 196 points $\mathbf{x}_i$ and each horizon $h$, a linear head on the final state of patch token $i$ predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[ \tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where $F_{ih}$ is the displacement of the surface point at $\mathbf{x}_i$, $o_{ih}$ its visibility, $s$ a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

**Target trajectory.** The flow samples the target only at the few patches it covers, too sparse to determine its rotation, so a dedicated linear head on the mean of the final summary states forecasts its rigid motion directly: for each horizon $h$, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\boldsymbol{\phi}}_h$,

trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \Big[ \|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \quad (3)$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t+\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t+\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge}) R(\tau_t)$, $d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. The rotation term matters for contact: a rotation error $\theta$ can displace a contact point at distance $r$ from the target's origin by up to $2r\sin(\theta/2)$. For example, a $5^\circ$ error at $r = 10\,\mathrm{cm}$ can shift the contact point by nearly 9 mm.

### 3.4. World–Action Learning and Inference

**Action stream.** The action stream generates the chunk by conditional flow matching [23]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command. In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With $S$, $M$, and $A$ denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in $X$ attend to tokens in $\mathcal{Y}$. The world stream never reads the actions. The action stream reads, in every layer, both the patch states shaped by scene-flow supervision and the summary states shaped by target-trajectory supervision.

**Training objective.** For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream $v_\psi$ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \quad (4)$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}, \quad (5)$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\mathrm{aux}}$ sums auxiliary appearance losses.

**Inference.** At deployment, FoveaWAM skips the forecast heads, as recent world–action models do [21, 54, 57]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary. On the robot, the latency of a query also includes tracking the target in the newest frame. Further implementation details are in the supplementary material.

## 4. Experiments

We test whether role-aware precision and low latency together improve manipulation when the scene keeps moving during inference. We first compare FoveaWAM with existing policies, each charged with its own latency (Sec. 4.2), then analyze which parts of its design account for the results (Sec. 4.3), and finally verify it on a real robot with tracked target poses (Sec. 4.4).

### 4.1. Experimental Setup

**OnTimeBench.** Unlike the evaluators of DOMINO [13] and PhysMani-Bench [55], which advance the simulation only after a policy has returned its prediction, OnTimeBench shares one physical clock among observation, scene motion, and action execution, so the scene keeps moving while a policy computes. It contains fourteen tasks in five families, built in MuJoCo/robosuite [41, 63]: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 4). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all OnTimeBench results on 1,400 held-out scenes (100 per task).

**Evaluation.** During each query, physics advances by the method's inference latency, drawn from its latency distribution measured in its native deployment setting at batch size 1 on one RTX 5090, the GPU of our robot. All methods share one executor, which executes each chunk from the command meant for the control-period boundary at which the chunk arrives. Besides this native setting, we add 150 ms of latency to every method and, as a reference without the cost of computation, pause physics during inference. We report the task-balanced success rate (SR) and the p95 time-to-action, the measured time from an observation to its first command; paired 95% intervals come from a bootstrap over held-out scenes. The supplementary material gives the full evaluation protocol.

**Baselines.** We compare with the reactive policies ACT [60] and Diffusion Policy [11] and the VLA $\pi_{0.5}$ [5]; the dynamic VLAs DynamicVLA [48] and PUMA [13]; the WAM

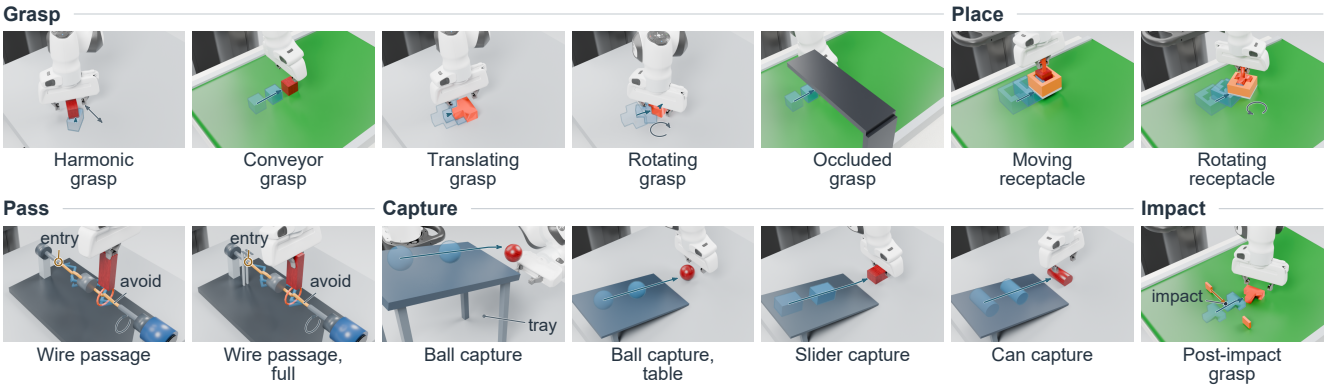


Figure 4. **OnTimeBench.** Fourteen tasks in five families, rendered from recorded simulator states with a common agent view. Translucent blue poses show selected history, fading with age; navy arrows follow recorded translational target motion. Local gray symbols denote rotation or oscillation qualitatively. An amber arrow shows the recorded striker approach; the *impact* label marks its contact location. Both Pass panels show passage: the wire is a target at entry and an obstacle along the path; Full additionally requires acquisition, exit and release. Occluded motion stays hidden. These native-expert examples illustrate task dynamics, not FoveaWAM outcomes; no future prediction is shown.

OpenWAM-$\alpha$ [43]; and the dynamic WAM DynamicWAM [27] (Table 1). Each baseline is trained on the same demonstrations with its official architecture and a task-adapted recipe. FoveaWAM receives a single agent-view RGB-D frame and a history of ground-truth SE(3) target poses from the simulator. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

**Implementation.** The transformer of FoveaWAM has 12 layers of width 1024 with 16 heads and 271.8M trainable parameters on top of the frozen MAE ViT-B/16 encoder. On OnTimeBench, it trains from random initialization with fused AdamW for up to 20 epochs. We evaluate the final checkpoint of the seed-0 run after 10 epochs (25,870 updates). Evaluation uses compiled BF16 and five Euler steps with TF32 disabled. Compilation and recorded-observation warm-up finish before physics starts. We then reset the history and inference seed.

## 4.2. Dynamic Manipulation Results

**Comparison with baselines.** With every method charged its own latency, FoveaWAM reaches 82.1% SR on held-out OnTimeBench scenes (Table 1), 20.7 points above the strongest baseline, PUMA (61.4%; 95% paired interval [18.4, 23.1]). Target-motion input and low latency are both needed for this result. Given the same SE(3) history, $\pi_{0.5}$ and DynamicWAM gain 4.3 and 0.6 points and reach 58.7% and 41.7%; ACT, the only faster method, reaches 41.6%. FoveaWAM and ACT are the only methods whose queries finish within one 50 ms control period, and every other baseline hands over its chunk at least two periods after capture. FoveaWAM has the highest or tied-highest SR on 13 of the 14 tasks (Table S4). Its margins over the best baseline are largest where a moving receptacle or the target's orientation decides success: 30 and 55 points on placement into translating and rotating receptacles, and 28 points on full wire passage.

Table 1. **Results on OnTimeBench.** Task-balanced SR (%) on 1,400 held-out scenes with each method's native latency, with 150 ms added, and with physics paused during inference. p95: time-to-action in each method's native deployment setting on one RTX 5090. History: inputs that carry past motion beyond the current frame. OpenWAM-$\alpha$ uses its official caching and compilation, and (off) disables both. PUMA is timed with its official three-camera input; the other baselines with two cameras; FoveaWAM runs compiled BF16 inference on one RGB-D view and ground-truth target poses from the simulator. Best and second best per column are highlighted.

| Method | History | | | SR (%) ↑ | | | p95 (ms) ↓ |
|---|---|---|---|---|---|---|---|
| | Wrist RGB | Flow | SE(3) | Native | +150 ms | Paused | |
| ACT [60] | | | | 41.6 | 24.6 | 40.6 | **5.8** |
| DP [11] | ✓ | | | 31.9 | 16.1 | 32.0 | 68.9 |
| $\pi_{0.5}$ [5] | | | | 54.4 | 28.1 | 66.6 | 59.6 |
| $\pi_{0.5}$+SE(3) | | | ✓ | 58.7 | 28.0 | 76.0 | 61.2 |
| DynamicVLA [48] | ✓ | | | 51.1 | 23.5 | 54.9 | 97.3 |
| PUMA [13] | | ✓ | | 61.4 | 31.1 | 53.1 | 60.6 |
| OpenWAM-$\alpha$ [43] | | | | 46.1 | 24.8 | 58.7 | 151.7 |
| OpenWAM-$\alpha$ (off) | | | | 15.8 | 10.4 | 58.4 | 466.3 |
| DynamicWAM [27] | | ✓ | | 41.1 | 20.6 | 49.6 | 149.6 |
| DynamicWAM+SE(3) | | ✓ | ✓ | 41.7 | 24.9 | 64.0 | 153.3 |
| FoveaWAM | | | ✓ | **82.1** | **34.1** | **90.3** | 15.1 |

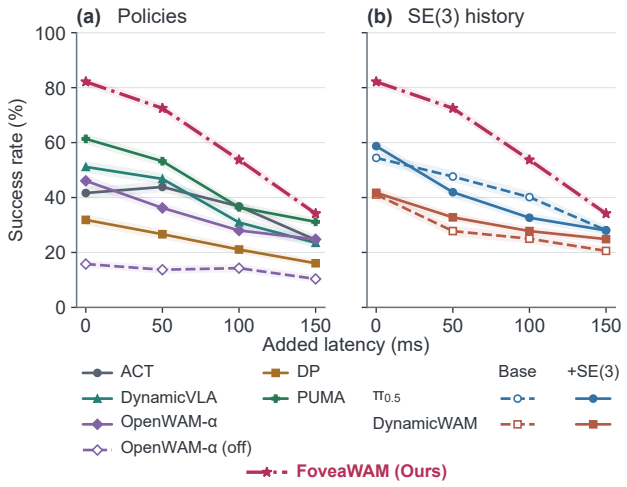


Figure 5. **Success under added latency.** Task-balanced SR on 1,400 held-out OnTimeBench scenes. Delay is added to each method's native latency, up to 150 ms, the largest step at which every method's chunk still covers the interval to its next chunk (Table S5 adds +200 ms). (a) Baseline deployments; off disables OpenWAM-$\alpha$ caching and compilation. (b) Base versus +SE(3)-history variants. PUMA is timed with its official three-camera input. FoveaWAM uses compiled BF16 (p95 15.1 ms). Bands: 95% task-stratified scene-bootstrap intervals.

**Effect of inference delay.** Latency alone can decide success. The two OpenWAM-$\alpha$ deployments run the same policy and agree with physics paused (58.7% and 58.4%), yet at native latency the one with a p95 of 151.7 ms reaches 46.1% and the one with 466.3 ms reaches 15.8%. Latency also erodes the value of motion input: the SE(3) history raises $\pi_{0.5}$ and DynamicWAM by 9.4 and 14.4 points with physics paused but by 4.3 and 0.6 at native latency. Charging latency therefore changes the ranking: $\pi_{0.5}$+SE(3), the strongest baseline with physics paused (76.0%), loses 17.3 points and falls behind PUMA. FoveaWAM leads in both settings, by 14.3 points over $\pi_{0.5}$+SE(3) and 20.7 over PUMA. Under added latency (Fig. 5), FoveaWAM remains the best method at every delay, ahead of the strongest baseline by 19.3 points at +50 ms, 13.6 at +100 ms, and 3.0 at +150 ms.

**Results on DOMINO.** To test the design beyond OnTimeBench, we train FoveaWAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [13] with its demonstrations and evaluate them with its native evaluator. As on OnTimeBench, FoveaWAM reads ground-truth target poses from the simulator. This evaluator pauses physics during inference, so FoveaWAM gains nothing from its low latency here; it still reaches the highest overall SR, 46.8%, 19.9 points above the strongest baseline, DynamicWAM (26.9%), and the highest manipulation score (Table 2). It leads on the 16 Track tasks, 55.5% against 31.9% for DynamicWAM; PUMA is strongest on the 8 Intercept tasks at 36.3%. More results are in the supplementary material.

Table 2. **DOMINO results** on 24 single-arm Franka clean-L1 tasks, with the same 2,400 test scenes per method (100 per task). Intercept/Track contain 8/16 tasks; All weights all 24 tasks equally. SR and MS use a 0–100 scale and retain failures. Best and second best are highlighted.

| Method | Intercept | Track | All ↑ | MS ↑ |
|---|---|---|---|---|
| $\pi_{0.5}$ | 23.8 | 8.1 | 13.3 | 27.8 |
| DynamicWAM | 16.8 | 31.9 | 26.9 | 41.4 |
| PUMA | **36.3** | 12.2 | 20.2 | 35.4 |
| FoveaWAM | 29.4 | **55.5** | **46.8** | **62.0** |

Table 3. **Analysis of role-aware modeling.** Task-balanced SR (%) of FoveaWAM on held-out OnTimeBench scenes and its change for each control ($\Delta$, points). Red marks drops and green marks gains, with shading scaled within each column. Rot./Ctx.: rotation-sensitive and context-active tasks (Table S2). Table S7 adds paused physics.

| | Success rate, all tasks | | Groups (Native) | |
|---|---|---|---|---|
| | Native | +150 ms | Rot. | Ctx. |
| **FoveaWAM** | **82.1** | **34.1** | **80.4** | **82.7** |
| *Precision allocation* | | | | |
| Uniform coarse flow | −4.5 | −0.5 | −8.6 | −4.6 |
| Uniform fine flow | +0.8 | −0.9 | −3.4 | −2.6 |
| w/o flow | −4.2 | −1.4 | −11.7 | −3.9 |
| *Target motion history* | | | | |
| w/o history | −19.6 | +0.2 | −24.0 | −2.3 |
| *Future supervision* | | | | |
| w/o future | −4.3 | −3.4 | −12.6 | −2.1 |
| w/o $\mathcal{L}_{\text{aux}}$ | −2.6 | −1.9 | −11.6 | +0.1 |

Table 4. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95: end-to-end time-to-action (ms) on one RTX 5090, including tracking for FoveaWAM. Best per column is highlighted, including ties.

| Method | Belt grasp | Box place | Turntable | Toy car grasp | Tennis ball | Ramp grasp | All ↑ | p95 (ms) ↓ |
|---|---|---|---|---|---|---|---|---|
| $\pi_{0.5}$ | — | — | — | — | — | — | — | — |
| DynamicWAM | — | — | — | — | — | — | — | — |
| ACT | — | — | — | — | — | — | — | — |
| FoveaWAM | — | — | — | — | — | — | 37.5 | — |

## 4.3. Analysis of Role-Aware Modeling

Six controls share the data, recipe, and executor of FoveaWAM (Table 3). The history control changes the input. The other five change only the supervision of prediction heads that are skipped at deployment, so they run the same

inference at the same latency as FoveaWAM, and any difference comes from what the model learned to forecast. We analyze SR at native latency, the condition FoveaWAM is designed for, overall and on the predefined rotation-sensitive and context-active task groups.

**Allocation of precision.** Role-aware supervision gives the target a precision that uniform supervision does not. At the same flow density, replacing the SE(3) trajectory with uniform flow reduces SR by 4.5 points overall and by 8.6 on rotation-sensitive tasks. Raising the uniform flow to a $56 \times 56$ grid, $16\times$ the points, costs [X.X]$\times$ the training time of FoveaWAM. It recovers the overall SR (+0.8) but still trails the role-aware model on both task groups, by 3.4 points on rotation-sensitive tasks and 2.6 on context-active tasks. One target trajectory and 196 flow points thus serve these tasks better than 3,136 uniform flow points, at a fraction of the training cost. The sparse flow is needed as well: removing it while keeping the SE(3) trajectory reduces SR by 4.2 points overall, by 3.9 on context-active tasks, and by 11.7 on rotation-sensitive tasks. The two forecasts are complementary, and FoveaWAM has the highest SR on rotation-sensitive tasks among all variants. All three controls keep the visual auxiliary objectives.

**Target motion history.** The target-motion history is the largest single factor. Keeping the current target pose but dropping its history, including the velocity and acceleration features, reduces SR by 19.6 points overall and by 24.0 on rotation-sensitive tasks. A single pose does not show how the target moves; the metric history supplies the motion that the commands must anticipate.

**Future supervision and auxiliary objectives.** Supervising the future, rather than only reconstructing the current target, is worth 4.3 points overall and 12.6 on rotation-sensitive tasks, and 3.4 points with 150 ms added, although the forecasting heads never run at deployment. The visual auxiliary objectives $\mathcal{L}_{\text{aux}}$ add 2.6 points overall and 11.6 on rotation-sensitive tasks. Every component of the role-aware design therefore contributes: removing any one of them lowers SR by 2.6 to 19.6 points overall and by 8.6 to 24.0 on rotation-sensitive tasks, where contact depends on the target's orientation.

### 4.4. Real-Robot Evaluation

**Setup.** The six tasks on a Flexiv Rizon 4s arm with a GN01 gripper include rolling tennis-ball interception (Fig. 6 and Table S8). Each method draws training data from a shared corpus of 600 teleoperated demonstrations. We plan 20 trials per task and method. FoveaWAM starts from a DROID-pretrained base [19]. In the planned evaluation, it uses RGB-D and online target pose estimates from the fixed camera only. The baselines ($\pi_{0.5}$, ACT, and DynamicWAM) use RGB from the fixed and wrist cameras.

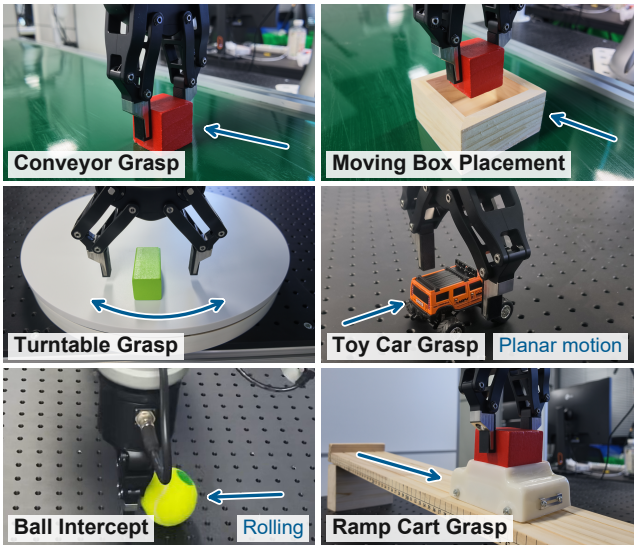


Figure 6. **Dynamic tasks on the real robot.** Straight arrows show example pre-contact motion of the target or moving box. The double-headed arc denotes turntable rotation. The tennis ball image is a demonstration frame.

**Results.** FoveaWAM succeeds in 37.5% of all trials, against [XX.X]% for [the strongest baseline] (Table 4). Its p95 latency, including tracking, is [XX] ms, [within one control period], whereas [name] needs [XX] ms. The margin is largest on [tasks]. [Whether the ranking of Table 1 carries over to real perception, and what tracking costs.] [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of FoveaWAM or a failure of a baseline.] The dominant failure of FoveaWAM is [failure mode]. Additional details are in the supplementary material.

## 5. Conclusion

FoveaWAM spends precision where contact depends on it: a metric SE(3) history and forecast for the target and a sparse 3D flow for the whole scene, with the forecasts used only as training supervision. When every method is charged its own latency, these choices let FoveaWAM act both precisely and quickly: it outperforms reactive policies, VLAs, and world–action models, including ones built for moving targets, on OnTimeBench and on six real-robot tasks at a p95 latency of 15.1 ms on OnTimeBench with compiled BF16 inference, and it also leads on DOMINO. The strongest baseline is PUMA under native latency and $\pi_{0.5}$+SE(3) in the paused column. At native latency on OnTimeBench, removing target pose history and removing future supervision reduce

success by 19.6 and 4.3 percentage points, respectively. We will release OnTimeBench so that future dynamic policies are evaluated with their latency charged.

**Limitations.** Deployment requires a specified rigid target and reliable online pose estimates. The target forecast predicts one future per horizon and ignores the effects of robot contact on target motion. Uncertainty-aware forecasts and contact-conditioned dynamics could address these limits. Independently moving context lies outside our task set and would need its own motion history, for example from scene flow.

## LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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# A. Method and Training Details

## A.1. Architecture and Training

On OnTimeBench, each training window holds eight consecutive query positions, each with its own causal history (Table S1).

Table S1. Hyperparameters of FoveaWAM on OnTimeBench and training clusters used across the project.

| Parameter | Value |
|---|---|
| *Model* | |
| Input image | $224 \times 224$ RGB-D, one view |
| Visual encoder | MAE ViT-B/16, frozen |
| History length | 8 steps (350 ms) |
| Forecast horizons | 8 (50 ms–400 ms) |
| Summary tokens | 16 |
| Layers | 12 |
| Width | 1024 |
| Heads | 16 |
| Feed-forward width | 4096 |
| Trainable parameters | 271.8M |
| *Action* | |
| Action dimension | 7 |
| Chunk length | 8 |
| Euler steps | 5 |
| Noise samples per query | 8 |
| *Training* | |
| Optimizer | Fused AdamW |
| Betas | $(0.9,\ 0.999)$ |
| Weight decay | $10^{-4}$ |
| Learning rate | $2 \times 10^{-4}$ |
| LR schedule | cosine, 5,174 warm-up updates |
| Batch size | 512 windows |
| Epochs | 10 (cap: 20) |
| Reported checkpoint | 25,870 updates |
| Gradient clipping | 0.1 |
| $\lambda_F, \lambda_P, \lambda_R, \lambda_{\text{aux}}$ | 0.1 |
| Training precision | BF16 compute, FP32 parameters |
| Training clusters | $64 \times$ RTX PRO 6000, $32 \times$ H100 |

**Scene tokens.** Each patch token of the frozen MAE ViT-B/16 [15] adds a learned view embedding and linear embeddings of its mean valid metric depth and of the base-frame position $\mathbf{x}_i$ of its back-projected center pixel; a patch without valid depth at its center receives no position embedding. Only the patch queries and keys receive 3D rotary embeddings from $\mathbf{x}_i$, and the same points are the source points of the sparse 3D flow.

**State tokens.** The 8 target-state tokens sample the offsets $-350, -300, \ldots, 0$ ms. Each is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$: slots 0–6 describe the seven intervals between consecutive offsets, with velocities and accelerations from finite differences over the elapsed times, and slot 7 holds the current pose in the base frame. Each dimension is standardized with training statistics, invalid dimensions are zeroed, and three validity flags (pose, interval, acceleration) are appended; a slot without any valid feature is masked out of attention. The robot tokens hold the end-effector position, rotation vector, and gripper opening at the same offsets, and 16 learned embeddings form the summary tokens.

**Heads and losses.** Each head is a linear layer on the final world-stream states, except the target reconstruction, a two-layer MLP. The pose head reads the mean summary state ($\mathcal{L}_P$, Eq. 3). The flow head gives each patch point's displacement and visibility logit at every horizon ($\mathcal{L}_F$, Eq. 2, with scale $s = 0.715$ m and visibility weight 1); valid but invisible points still supervise their displacement. Flow labels follow the rendered surface point at each patch center through the ground-truth rigid motion of its MuJoCo geometry; a future point is visible if it projects into the image onto the same geometry within a depth tolerance of $\max(1\,\text{mm},\ 0.001z)$. $\mathcal{L}_{\text{aux}}$ sums four losses: squared errors on future metric depth (valid pixels), on the RGB of surfaces that move by more than 1 mm, and on layer-normalized DINOv3 features [34], and an absolute error on the current target's RGB region reconstructed from the mean summary state.

**Action stream.** Each action token embeds a seven-dimensional command: a base-frame translation increment, a left-relative rotation vector, and a gripper command, relative to the measured end-effector pose at application. Command $j$ of query $t$ is labeled with the demonstration command issued at $\tau_t + j\Delta_a$, since $d_{\text{task}}$ applies only at execution (Eq. (S1)), and commands are standardized over the training split. At deployment, the gripper changes state only when its command leaves a deadband of 0.5 in Ball capture and 0.1 in every other task, including Ball capture, table.

## A.2. Pretraining on DROID and Fine-Tuning

The real-robot base trains on DROID [19] (15 Hz) with the first exterior camera and its depth, using the 3D point tracks and calibrations released by PointWorld [17]. Its 42,087 usable episodes yield 598,889 anchor windows, 60,149 of them for validation; each anchor expands into 8 query positions as on OnTimeBench, and only the anchor carries flow and DINOv3 labels. Each patch follows one PointWorld track that starts inside it, preferring the one with the largest consistent motion so that small moving objects are covered; 47% of the point–horizon slots carry a label.

DROID provides no object poses, so the base omits the target-state tokens and pose head. The losses for actions, sparse 3D flow, and future DINOv3 features have weights 1, 0.1, and 0.1, respectively. The flow loss has no visibility term. We cap pretraining at 20 epochs (21,060 optimizer updates)

with AdamW and a global batch size of 512. The backbone uses FP32 and the frozen visual encoder uses BF16.

Each real-robot policy starts from a DROID-pretrained base and is fine-tuned on its task's demonstrations, with target-state tokens and new prediction heads initialized from scratch. The fine-tuning budget is 20,000 optimizer updates per task with a global batch of 32. AdamW uses 500 warm-up updates followed by cosine decay, with peak learning rates of $3 \times 10^{-5}$ and $2 \times 10^{-5}$ for the 12- and 24-layer bases, respectively.

# B. OnTimeBench and Evaluation Protocol

## B.1. Benchmark Design

The tasks run in MuJoCo 3.11.0 and robosuite 1.5.2 [41, 63] with a Franka Panda arm and a parallel gripper under robosuite's operational-space pose controller, with a control period of 50 ms and a physics step of 2 ms. A fixed third-person camera with a $40^\circ$ vertical field of view renders $224 \times 224$ RGB-D images. An action is an end-effector translation and rotation-vector increment in the base frame with a gripper command, clipped to 0.10 m per translation axis and 0.5 rad per rotation component. Table S2 lists the fourteen tasks.

## B.2. Scenarios, Demonstrations, and Splits

Each scene samples the target's initial state and motion parameters uniformly within task-specific ranges. Table S2 summarizes target motion. Other sampled parameters include harmonic amplitudes of up to 35 mm at 0.25–0.7 Hz, yaw rates of up to 1.2 rad/s, and ramp slopes of 5–15°. In an occluded scene, the target is hidden for 0.5–3.0 s and then reappears. Only physically feasible scenes are kept, and evaluation replays the stored scenes.

The expert is a task-specific closed-loop state machine with privileged simulator state that anticipates the target's motion, for example by predicting the interception point in the capture tasks. A demonstration is kept if the expert succeeds, a replay of its actions reproduces the episode, and its path and jerk pass quality checks. The training pool holds 11,200 demonstrations, 800 per task, recorded at 20 Hz with RGB-D, robot states, commands, and simulator ground truth, and 350 further episodes form the validation split; policies receive only the inputs of Sec. 3.

We select configurations on 1,400 development scenes (100 per task) and report all OnTimeBench results on 1,400 held-out scenes shared by all methods, generated with the same ranges from seeds used in no training, validation, or development episode.

## B.3. Real-Time Evaluation and Metrics

**Executor.** All methods share one executor on the 50 ms control clock. Command $j$ of query $t$ is meant for $\tau_t + j\Delta_a$, and its execution time (Sec. 3.1) is

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + j\Delta_a + d_{\mathrm{task}}, \tag{S1}$$

where $d_{\mathrm{task}} = 50$ ms for the post-impact grasp, which applies each command 50 ms after it is issued, and $d_{\mathrm{task}} = 0$ otherwise. As in RTC [7], latency counts in whole control periods: a query with total latency $L_t$ hands over its chunk at the first boundary at or after it ends, $k_t = \lceil L_t/\Delta_a \rceil$ periods after capture, and execution continues from command $k_t$, the one meant for that boundary. The policy then observes again and starts the next query; meanwhile the previous chunk continues, so a policy replans as often as its latency allows. A chunk holds the method's native number $H_m$ of commands (Table S3) and is dropped if $k_t \geq H_m$. Before the first chunk arrives and whenever a chunk runs out, the controller holds the current arm target and gripper command; no command is repeated. The executor runs asynchronously, as in the streaming of DynamicVLA; no method, including FoveaWAM, inpaints across chunks as in RTC.

**Protocols.** In real time, physics advances during each query by the method's latency, drawn per query from its distribution measured on one RTX 5090, while the rollouts themselves run on RTX PRO 6000. With paused physics, inference takes no physics time and a policy replans every 50 ms; this removes the wait for inference but also raises the feedback rate, so it is a reference rather than an upper bound. The latency sweep adds 0 to 200 ms in steps of 50 ms after inference in real time.

**Latency measurement.** We time each method alone on one RTX 5090 at batch size 1 in each baseline's release precision and in compiled BF16 for FoveaWAM, over three rounds of 30 warm-up and 300 measured calls on recorded observations with the required causal history. A timed query starts from the newest observation and ends when its first command is available. It includes all computation that needs the newest frame, such as preprocessing, image encoding, the online flow and motion descriptors of PUMA and DynamicWAM, sampling, and action decoding, and on the robot also tracking. The forecast heads of FoveaWAM are skipped at deployment and excluded from time-to-action. A query misses its deadline when this time-to-action exceeds 50 ms, and the action age of a command is its execution time minus the capture time. Appendix C.2 reports the measured values.

**Metrics.** SR weights the 14 tasks equally. The position error at first contact $e_p$ is the distance of the end effector from the expert's contact frame on the target; in the placement tasks, which start with the object held, it uses the contact of the held object with the receptacle and the expert's placement

Table S2. **The fourteen OnTimeBench tasks.** The target is the body whose pose the policy receives (Sec. 3.1). For placement it is the receptacle. For passage it is the wire entry, and the rest of the wire is context. Motion gives linear speed in m/s or the wire's angular speed in degrees per second. For each capture task, we estimate speed from the two samples 50 ms apart that bracket the first recorded ramp exit in each of 100 held-out fixed-robot reference trajectories. A grasp requires bilateral fingerpad contact, a lift of at least 0.05 m, and a continuous 0.5 s hold. An episode fails at unsafe contact, leaving the workspace, a drop, or the time limit. In the group column (Table 3), R marks tasks where the target rotates during the lead and its orientation matters at contact. C marks tasks with an obstacle clearance or a driver such as a conveyor, ramp, or striker.

| Task | Code | Target | Motion range | Success | Limit (s) | Group |
|---|---|---|---|---|---|---|
| Harmonic grasp | `grasp` | block | 0.03–0.20 | grasp | 8 | R |
| Conveyor grasp | `t1_belt` | part | 0.04–0.10 | grasp | 8 | C |
| Translating grasp | `t1_xyz` | part | 0.04–0.15 | grasp | 8 | – |
| Rotating grasp | `t1_so3` | part | 0.04–0.15 | grasp | 8 | R |
| Occluded grasp | `t1_occ` | part | 0.04–0.10 | grasp without canopy contact | 8 | C |
| Moving receptacle | `t2_trans` | receptacle | 0.03–0.06 | held object at rest inside | 10 | – |
| Rotating receptacle | `t2_se3` | T-shaped receptacle | 0.03–0.06 | held T-part at rest inside | 10 | R |
| Wire passage | `t3_phase` | wire entry | 2.5–7.5°/s | align within 30 mm, pass without contact | 21 | R, C |
| Full wire passage | `t3_full` | wire entry | 2.5–7.5°/s | grasp wand, pass, exit, release | 32 | R, C |
| Ball capture | `t4_sphere` | ball | 0.67–1.14 | grasp before the tray | 6 | C |
| Ball capture on table | `t4_sphere_tabletop` | ball | 0.59–1.03 | grasp while rolling | 6 | C |
| Slider capture | `t4_slider` | slider | 0.55–1.12 | grasp while sliding | 6 | C |
| Can capture | `t4_can` | can | 0.59–0.97 | grasp before tipping | 6 | R, C |
| Post-impact grasp | `t5_replan` | part | 0.12–0.14 | grasp along its axis | 6 | R, C |

pose. $e_p$ averages over the 13 tasks with a contact, excluding `t3_phase`, and over the episodes that make contact; failures without contact remain in the SR denominator. Paired 95% intervals bootstrap held-out scenes within each task, jointly across methods.

### B.4. Baselines

Table S3. **Baseline training.** The training budget gives optimizer updates and global batch size. $H$ is the number of commands returned per query. The parameter counts include trainable parameters only. DOM ckpt. denotes the released DynamicVLA checkpoint trained on DOM.

| Method | Initialization | Updates×batch | $H$ | Params |
|---|---|---|---|---|
| ACT [60] | scratch | $50{,}000 \times 32$ | 100 | 83.9M |
| DP [11] | scratch | $50{,}000 \times 32$ | 15 | 288.7M |
| $\pi_{0.5}$ [5] | `pi05_base` | $30{,}000 \times 256$ | 10 | 3,353.4M |
| $\pi_{0.5}$+SE(3) | `pi05_base` | $30{,}000 \times 256$ | 10 | 3,357.7M |
| DynamicVLA [48] | DOM ckpt. | $50{,}000 \times 44$ | 20 | 99.1M |
| PUMA [13] | official | $40{,}000 \times 64$ | 16 | 4,516.5M |
| OpenWAM-$\alpha$ [43] | official | $34{,}490 \times 192$ | 32 | 6,021.2M |
| DynamicWAM [27] | official | $40{,}000 \times 128$ | 16 | 988.8M |
| DynamicWAM+SE(3) | official | $40{,}000 \times 128$ | 16 | 989.4M |
| FoveaWAM | scratch | $25{,}870 \times 512$ | 8 | 271.8M |

**Common conditions.** Each baseline keeps its official architecture, chunk length, initialization, and training recipe (Table S3), trains on the same 11,200 demonstrations mapped to its action space through the same Cartesian controller, and is configured on development scenes. Rollouts use the agent view each baseline was trained with, any wrist slot masked, and the histories of Table 1; masking the wrist view changes p95 latency by at most 3.8 ms in the seven configurations timed both ways. DP uses its authors' real-robot DDIM sampler with 16 steps, and OpenWAM-$\alpha$ runs with video decoding disabled and is timed with and without its official cache and compilation.

**SE(3)-augmented baselines.** $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.1 from the same pose source, with the same validity flags. A two-layer MLP with a zero-initialized output layer maps them to the model's width; for $\pi_{0.5}$ they follow the image tokens in the prefix, where the action expert attends to them, and for DynamicWAM they join its kinematic descriptors. Training follows the base model's recipe.

## C. Additional Simulation Results

### C.1. Per-Task Results

Table S4 uses the Native real-time protocol with +0 ms extra delay. FoveaWAM reaches 93% on moving-receptacle placement and 66% on rotating-receptacle placement, compared with 63% and 11% for $\pi_{0.5}$+SE(3). On full wire passage, FoveaWAM reaches 40% versus 12% for DynamicVLA. It ties the strongest baseline on phase-only wire passage, tabletop ball capture and can capture. The largest recorded failure classes for FoveaWAM are rim impulse violations (160 of 1,400 scenes) and wire contacts (55 scenes).

### C.2. Latency

Table S5 lists all eleven configurations at five added latencies; Fig. 5 plots the points from 0 to 150 ms. At +200 ms the 8-

Table S4. **Per-task held-out success** (%) on OnTimeBench for every method of Table 1, Native real-time protocol: each method's measured inference latency plus **+0 ms extra delay**, with physics continuing during inference; 100 paired scenes per task. FoveaWAM uses compiled BF16 C (p95 15.1 ms). DP: Diffusion Policy. DVLA: DynamicVLA. OWAM: OpenWAM-$\alpha$ with official cache/compile; OW-off: both disabled. DWAM: DynamicWAM. +: with our SE(3) history tokens. $\Delta$: FoveaWAM minus the strongest other method on that task; ties follow table order. The task-balanced row compares against the strongest overall baseline, PUMA. Brackets give pointwise 95% paired scene-bootstrap intervals, conditional on the selected comparator and checkpoint (100,000 resamples; no selection adjustment). The last two rows give first-contact position error $e_p$ (mm), averaged over the 13 contact tasks, and the number $n$ of episodes with a contact (Appendix B.3); –: no complete 13-task mean. $\Delta$ shading is blue for higher SR and orange for lower, saturating at 15 points. Blue/bold and light blue mark the best and second-best displayed values.

| Task | ACT | DP | $\pi_{0.5}$ | $\pi_{0.5}$+ | DVLA | PUMA | OWAM | OW-off | DWAM | DWAM+ | **Ours** | $\Delta$ [95% CI] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Harmonic grasp | 49.0 | 31.0 | 51.0 | 84.0 | 39.0 | 85.0 | 41.0 | 1.0 | 50.0 | 47.0 | **99.0** | +14.0 [7.0, 22.0] |
| Conveyor grasp | 45.0 | 45.0 | 96.0 | 62.0 | 89.0 | 89.0 | 62.0 | 0.0 | 59.0 | 60.0 | **100.0** | +4.0 [1.0, 8.0] |
| Translating grasp | 14.0 | 33.0 | 59.0 | 42.0 | 61.0 | 51.0 | 53.0 | 0.0 | 53.0 | 18.0 | **77.0** | +16.0 [4.0, 28.0] |
| Rotating grasp | 24.0 | 20.0 | 32.0 | 53.0 | 26.0 | 67.0 | 42.0 | 3.0 | 30.0 | 35.0 | **71.0** | +4.0 [-8.0, 16.0] |
| Occluded grasp | 30.0 | 28.0 | 37.0 | 60.0 | 80.0 | 84.0 | 34.0 | 0.0 | 44.0 | 38.0 | **100.0** | +16.0 [9.0, 23.0] |
| Moving receptacle | 26.0 | 24.0 | 34.0 | 63.0 | 41.0 | 24.0 | 27.0 | 0.0 | 8.0 | 6.0 | **93.0** | +30.0 [21.0, 40.0] |
| Rotating receptacle | 0.0 | 0.0 | 1.0 | 11.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | **66.0** | +55.0 [45.0, 65.0] |
| Wire passage | 99.0 | 99.0 | **100.0** | 89.0 | 99.0 | 87.0 | 50.0 | 58.0 | 23.0 | 78.0 | **100.0** | +0.0 [0.0, 0.0] |
| Wire passage, full | 0.0 | 3.0 | 1.0 | 0.0 | 12.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | **40.0** | +28.0 [16.0, 40.0] |
| Ball capture | 36.0 | 16.0 | **46.0** | 29.0 | 41.0 | 39.0 | 45.0 | 13.0 | 24.0 | 20.0 | 38.0 | -8.0 [-18.0, 2.0] |
| Ball capture, table | 71.0 | 81.0 | **100.0** | **100.0** | 88.0 | **100.0** | 99.0 | 88.0 | 98.0 | **100.0** | **100.0** | +0.0 [0.0, 0.0] |
| Slider capture | 56.0 | 30.0 | 58.0 | 62.0 | 28.0 | 76.0 | 71.0 | 31.0 | 47.0 | 63.0 | **79.0** | +3.0 [-5.0, 12.0] |
| Can capture | 91.0 | 31.0 | 98.0 | 99.0 | 72.0 | **100.0** | **100.0** | 5.0 | 96.0 | **100.0** | **100.0** | +0.0 [0.0, 0.0] |
| Post-impact grasp | 42.0 | 5.0 | 49.0 | 68.0 | 39.0 | 57.0 | 21.0 | 22.0 | 43.0 | 19.0 | **87.0** | +19.0 [7.0, 31.0] |
| Task-balanced | 41.6 | 31.9 | 54.4 | 58.7 | 51.1 | 61.4 | 46.1 | 15.8 | 41.1 | 41.7 | **82.1** | +20.8 [18.4, 23.1] |
| $e_p$ (mm) | 32.2 | 36.1 | 31.3 | 34.8 | 30.0 | 27.2 | 31.8 | – | 31.2 | 36.1 | **22.0** | – |
| $n$ | 1110 | 946 | 1237 | 1191 | 1204 | 1272 | 1115 | 493 | 1193 | 1029 | 1285 | – |

command chunk of FoveaWAM and the 10-command chunk of $\pi_{0.5}$ no longer cover the interval until the next chunk arrives: three and four commands remain for five and six control periods, and the controller holds its last target in between. That column therefore measures chunk length rather than the response to delay.

Table S5. **Success versus added latency** on 1,400 held-out OnTimeBench scenes per point (SR in %, real-time protocol). The 0 and 150 ms columns equal the Native and +150 ms columns of Table 1. Blue/bold and light blue mark the best and second-best values per column.

| | Added latency (ms) | | | | |
|---|---|---|---|---|---|
| Method | 0 | 50 | 100 | 150 | 200 |
| ACT | 41.6 | 43.9 | 36.8 | 24.6 | 16.6 |
| Diffusion Policy | 31.9 | 26.6 | 21.1 | 16.1 | 14.4 |
| $\pi_{0.5}$ | 54.4 | 47.6 | 40.1 | 28.1 | 13.6 |
| $\pi_{0.5}$ + SE(3) | 58.7 | 41.9 | 32.6 | 28.0 | 14.9 |
| DynamicVLA | 51.1 | 46.8 | 30.9 | 23.5 | 18.0 |
| PUMA | 61.4 | 53.2 | 36.4 | 31.1 | **27.2** |
| OpenWAM-$\alpha$ | 46.1 | 36.2 | 28.0 | 24.8 | 22.1 |
| OpenWAM-$\alpha$ (off) | 15.8 | 13.7 | 14.3 | 10.4 | 9.5 |
| DynamicWAM | 41.1 | 27.8 | 25.0 | 20.6 | 21.0 |
| DynamicWAM + SE(3) | 41.7 | 32.8 | 27.8 | 24.9 | 22.9 |
| FoveaWAM | **82.1** | **72.5** | **53.7** | **34.1** | 12.8 |

**Timing.** Table S6 reports replay timings by stage for the development checkpoint at 20,696 updates.

Table S6. **Latency breakdown** of one query of compiled BF16 FoveaWAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot. The policy total uses 900 measured queries and excludes pose tracking; magenta dashes mark pending measurements.

| Stage | p50 | p95 |
|---|---|---|
| Pose tracking (per frame, robot) | — | — |
| Observation encoding + H2D | 4.83 | 5.28 |
| World stream (once per query) | 3.93 | 4.22 |
| Action stream (5 Euler steps) | 4.27 | 4.41 |
| Policy time-to-action (total) | 14.66 | 15.65 |
| Training-only readouts (for reference) | 0.37 | 0.41 |
| Action age at execution (robot) | — | — |
| Queries over 50 ms (robot, %) | — | |

## C.3. Results on DOMINO

We use the 24 single-arm Franka tasks of the clean level-1 setting of DOMINO [13] with its demonstrations, scenes, and native evaluator. The manipulation score (MS) scales route completion by penalties for leaving the workspace or view and for collisions. FoveaWAM reads the ground-truth pose of the instructed target from the simulator and masks slots without a valid pose (Appendix A.1). The RGB-based

baselines do not receive this pose.

**Data and adaptation.** The data split has 1,200 training and 240 validation demonstrations. Policies execute absolute joint-position targets with DOMINO's native controller. Their state and action projections use its eight-dimensional joint space, with normalization fitted to the training data. The official sampler of DynamicWAM skips demonstrations too short for its windows, which leaves it 1,054 training and 210 validation demonstrations and none for two tasks; all 24 tasks remain in the test set. DynamicWAM trains for 40,000 updates at global batch 128. $\pi_{0.5}$ starts from its official base, trains for 30,000 updates at global batch 256, and is evaluated with its final EMA. PUMA keeps its published weights and frozen teacher and trains at global batch 64 with early stopping on validation action MSE, which selected update 6,000. For development, FoveaWAM trains from scratch with a global batch of 512 and stops after 1,575 updates. The lowest validation loss selects the checkpoint at update 1,365. Development rollouts select three executed actions per query.

### C.4. Ablation Controls

The six controls of Sec. 4.3 (Table 3) share the data, training recipe, and executor. Each is evaluated at the final checkpoint of its run. Removing a loss also removes its head. The control without target history removes slots 0–6 and keeps slot 7, the current pose (Appendix A.1). The uniform fine flow control removes $\mathcal{L}_P$ and the pose head. It supervises flow at $4\times4$ points in each patch, forming a $56\times56$ grid with labels from the same ground-truth rigid motion. Table S7 reports the controls in all three settings, including paused physics.

Table S7. **Ablation controls in all three settings.** Task-balanced SR (%) of FoveaWAM on held-out OnTimeBench scenes and its change for each control (Δ, points). The Native, +150 ms, Rot., and Ctx. columns equal Table 3; Paused adds the paused-physics protocol. Red marks drops and green marks gains, with shading scaled within each column. Rot./Ctx.: rotation-sensitive and context-active tasks (Table S2).

| | Success rate, all tasks | | | Groups (Native) | |
|---|---|---|---|---|---|
| | Paused | Native | +150 ms | Rot. | Ctx. |
| **FoveaWAM** | **90.3** | **82.1** | **34.1** | **80.4** | **82.7** |
| *Precision allocation* | | | | | |
| Uniform coarse flow | +4.4 | −4.5 | −0.5 | −8.6 | −4.6 |
| Uniform fine flow | +1.3 | +0.8 | −0.9 | −3.4 | −2.6 |
| w/o flow | +3.1 | −4.2 | −1.4 | −11.7 | −3.9 |
| *Target motion history* | | | | | |
| w/o history | −19.5 | −19.6 | +0.2 | −24.0 | −2.3 |
| *Future supervision* | | | | | |
| w/o future | +0.2 | −4.3 | −3.4 | −12.6 | −2.1 |
| w/o $\mathcal{L}_{\text{aux}}$ | −3.8 | −2.6 | −1.9 | −11.6 | +0.1 |

## D. Real-Robot Experiments

### D.1. Platform and Tasks

The robot is a Flexiv Rizon 4s arm with a GN01 gripper. Policies send absolute Cartesian tool-center-point poses and gripper targets at 20 Hz. Demonstration collection logs robot control at 100 Hz. The setup uses two RealSense D435 cameras, one fixed and one mounted on the wrist, both recording RGB at $640\times480$ and 60 Hz. We calibrate the fixed camera to the robot base with a ChArUco board. In the planned evaluation, FoveaWAM uses RGB-D frames and target poses from the fixed camera. Baselines use RGB from both cameras. The task devices are installed on this platform one at a time (Fig. 6; Table S8).

Table S8. **Real-robot tasks.** A grasp succeeds when the robot holds the target stably and lifts it clear of its support. The time limit is 30 s for every task.

| Task | Target motion | Success |
|---|---|---|
| Conveyor Grasp | Block on a conveyor belt | Grasp |
| Moving Box Placement | Open box on a conveyor belt | Block stays in box |
| Turntable Grasp | Rectangular block on a turntable | Grasp across short sides |
| Toy Car Grasp | Toy car moving across the table | Grasp |
| Tennis Ball Interception | Tennis ball rolls with randomized initial speed and heading offset | Stop or deflect with the gripper |
| Ramp Cart Grasp | Block carried by a cart on an incline | Grasp |

### D.2. Pose Tracking and Labels

**Offline labels.** We label the recorded fixed view with masks from Grounding DINO [24] and SAM 2.1 [31]. FoundationPose [44] tracks the metric object mesh forward and backward using depth from FoundationStereo [45]. We check agreement between directions and use tool-center-point kinematics to fill occluded frames during rigid grasps. We retain 595 demonstrations with valid target poses on 97.1% of their frames. Tennis-ball labels use translation only. For sparse scene flow, we lift CoTracker3 [18] tracks to the robot base frame using stereo depth. Future slots are sampled at the 50–400 ms prediction horizons by interpolating valid observations.

**Online pose estimation.** We estimate target poses from the fixed camera's aligned RGB-D frames and transform them to the robot base frame using the camera-to-base calibration. The policy samples these estimates at 20 Hz to build its target-state history.

**Figure placeholder: Real-robot failures**

**(a) Failures by category.** One stacked bar per method. **(b) Representative trials.** Show unretouched fixed-view frames and mark the events that determine the outcomes.

Figure S1. **Real-robot failures.** (a) Failure counts by category and method. (b) Frames from representative trials. [Tasks, methods, and outcomes of the shown trials.]

### D.3. Training and Evaluation Protocol

The shared corpus contains 600 teleoperated demonstrations, with 101 for Turntable Grasp, 99 for Tennis Ball Interception, and 100 for each other task. ACT trains one policy per task for 50,000 updates. DynamicWAM fine-tunes one policy across the six tasks for 20,000 updates. Its deployment configuration uses asynchronous 16-action chunks, requests a new chunk with four commands (200 ms) remaining, and blends overlapping commands. Appendix A.2 describes fine-tuning for FoveaWAM.

We run 20 trials for each task and method in separate groups. An operator resets the scene before each trial and records its outcome. We count timeouts, operator interventions, and safety stops as failures. Table S6 reports robot timing.

### D.4. Results and Failure Analysis

Table S9. **Real-robot success with intervals.** Success rates (%) pooled over the trials of each method, with 95% Wilson intervals. Table 4 gives per-task rates.

| Method | All [95% CI] |
|---|---|
| $\pi_{0.5}$ | — |
| DynamicWAM | — |
| ACT | — |
| FoveaWAM | — |

Table S9 gives pooled success rates and intervals. Figure S1 groups failures by observable events, including late or misaligned grasps, slips or drops, collisions or safety stops, timeouts, and loss of target tracking. [The dominant failures by method and the evidence for a timing error.]